

ONCOR ELECTRIC DELIVERY SYSTEM

2015 Analysis of System Losses

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Second Draft
1/17/17

Prepared by:



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January 17, 2017

Mr. _____
Title
Oncor Electric Delivery System
Street
City, ST Zip

RE: 2015 LOSS ANALYSIS

Dear Mr. _____:

Transmitted herewith are the results of the 2015 Analysis of System Losses for the Oncor Electric Delivery System's (Oncor) power system. Our analysis develops cumulative expansion factors (loss factors) for both demand (peak/kW) and energy (average/kWh) losses by discrete voltage levels applicable to metered sales data. Table 1 of the Executive Summary presents the results and appropriate loss factors to apply to metered load research or sales data for adjustment to system input.

On behalf of MAC, we appreciate the opportunity to assist you in performing the loss analysis contained herein. The level of detailed load research and sales data by voltage level, coupled with a summary of power flow data and power system model, forms the foundation for determining reasonable and representative power losses on the Oncor system. Our review of these data and calculated loss results support the proposed loss factors as presented herein for your use in various cost of service, rate studies, and demand analyses.

Should you require any additional information, please let us know at your earliest convenience.

Sincerely,

Paul M. Normand
Principal

Enclosure
PMN/rjp

Oncor Electric Delivery System 2015 Analysis of System Losses

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Appendix A – Results of Oncor Electric Delivery System 2015 Loss Analysis

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Oncor Electric Delivery System 2015 Analysis of System Losses

1.0 EXECUTIVE SUMMARY

This report presents Oncor Electric Delivery System's (Oncor) 2015 Analysis of System Losses for the power systems as performed by Management Applications Consulting, Inc. (MAC). The study developed separate demand (kW) and energy (kWh) loss factors for each voltage level of delivery service in the power system for Oncor. The cumulative loss factor results by voltage level, as presented herein, can be used to adjust metered kW and kWh sales data for losses in performing cost of service studies, determining voltage discounts, and other analyses which may require a loss adjustment.

The procedures used in the overall loss study emphasized the use of "in house" resources where possible. To this end, extensive use was made of the Company's ERCOT peak hour MW flow data and transformer plant investments in the model. In addition, measured and estimated load data provided a means of calculating reasonable estimates of losses by using a "top-down" and "bottom-up" procedure. In the "top-down" approach, losses through and including distribution substations, were calculated along with conductor and transformer loss estimates, and metered sales.

Our analyses began with system MW loads and losses at the input into the distribution substation system are known with reasonable accuracy. Estimated and actual Company load data provided the starting point for performing a "bottom-up" approach for calculating the remaining distribution losses. Basically, this "bottom-up" approach develops line loadings by first determining loads and losses at each level beginning at a customer's meter service entrance and then going through secondary lines, line transformers, primary lines and finally distribution substation which is the input level for this study. These distribution system loads and associated losses are then compared to the initial calculated input into Distribution Substation loadings for reasonableness prior to finalizing the loss factors. An overview of the loss study is shown on Figure 1.

The load research data provided the starting point for performing a "bottom-up" approach for estimating the remaining distribution losses. These distribution system loads and associated losses are then compared to the initial calculated input into Distribution Substation loadings for reasonableness prior to finalizing the loss factors.

Table 1, below, provides the final results from Appendix A for the 2015 calendar year. Exhibits 8 and 9 of Appendix A present a more detailed analysis of the final calculated summary results of losses by voltage segments and delivery service level in the power system. These Table 1 cumulative loss expansion factors are applicable only to metered sales at the point of receipt for adjustment to the power system's input level which is substations in this study.



Oncor Electric Delivery System 2015 Analysis of System Losses

TABLE 1
Loss Factor Multipliers at Metered Sales Level, Calendar Year 2015

<u>Voltage Level of Service</u>	<u>Oncor Delivery System</u>
<u>Demand (kW)</u>	
Transmission	N/A
Primary Lines	1.02019
Secondary	1.03885
<u>Energy (kWh)</u>	
Transmission	N/A
Primary Lines	1.01506
Secondary	1.03717
Losses –	MWH
Net System Input ¹	MW
	3.25%
	3.56%
Losses –	MWH
Net System Output ²	MW
	3.36%
	3.69%
Composite System Loss Factors at Metered Sales Level	
System	MWH
	1.03362
	MW
	1.03688

The net system input shown in Table 1 represents the MWh losses of 3.25% for the total Oncor load using calculated losses divided by the associated input energy to the system. The 3.36% represents the same losses using system output instead of input as a reference. The net system output reference shown in Table 1 represents MWh losses of 3.36% and MW losses of 3.69%. These results use the appropriate total losses for each but are divided by system output or sales. These calculations are all based on the data and results shown on Exhibits 1, 7 and 9 of the study.

Due to the very nature of losses being primarily a function of equipment loading levels for a peak load hour, the loss factor derivations for any voltage level must consider both the load at that level plus the loads from lower voltages and their associated losses. As a result, cumulative losses on losses equates to additional load at higher levels along with future changes (+ or –) in loads throughout the power system. It is therefore important to recognize that losses are multiplicative in nature (future) and not additive (test year only) for all future years to ensure total recovery based on prospective fixed loss factors for each service voltage.

¹ Net system input equals firm sales plus losses. See Appendix A, Exhibit 1, for their calculations.

² Net system output uses losses divided by output or sales data as a reference (excludes losses).

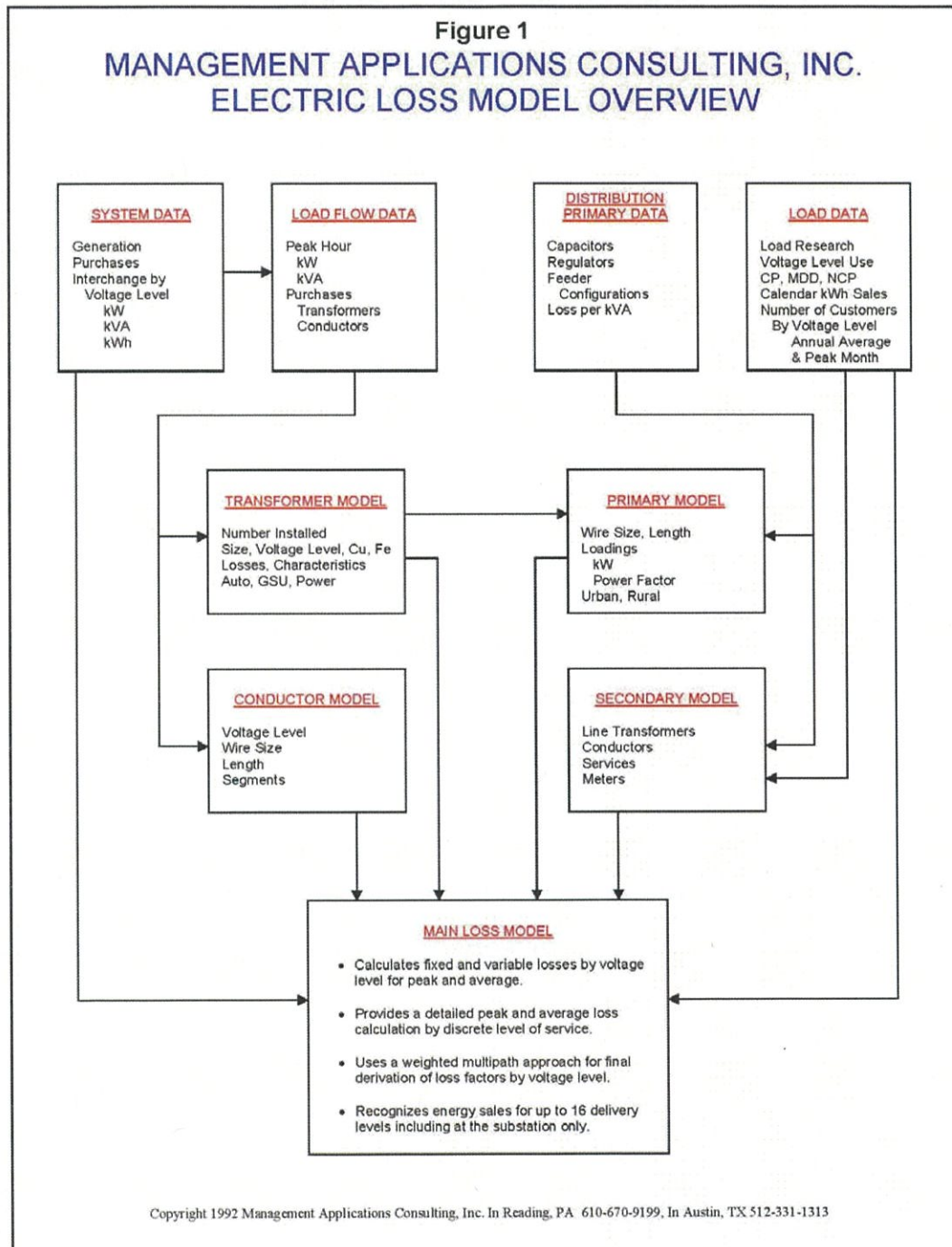
Oncor Electric Delivery System 2015 Analysis of System Losses

The derivation of the cumulative loss factors shown in Table 1 have been detailed for all electrical facilities in Exhibit 9, page 1 for demand and page 2 for energy. Beginning on line 1 of page 1 (demand) under the secondary column, metered sales are adjusted for service losses on lines 3 and 4. This new total load (with losses) becomes the load amount for the next higher facilities of secondary conductors and their loss calculations. This process is repeated for all the installed facilities until the secondary sales are at the input level (line 45). The final loss factor for all delivery voltages using this same process is shown on line 46 and Table 1 for demand. This procedure is repeated in Exhibit 9, page 2, for the energy loss factors.

The loss factor calculation is simply the input required (line 45) divided by the metered sales (line 43).

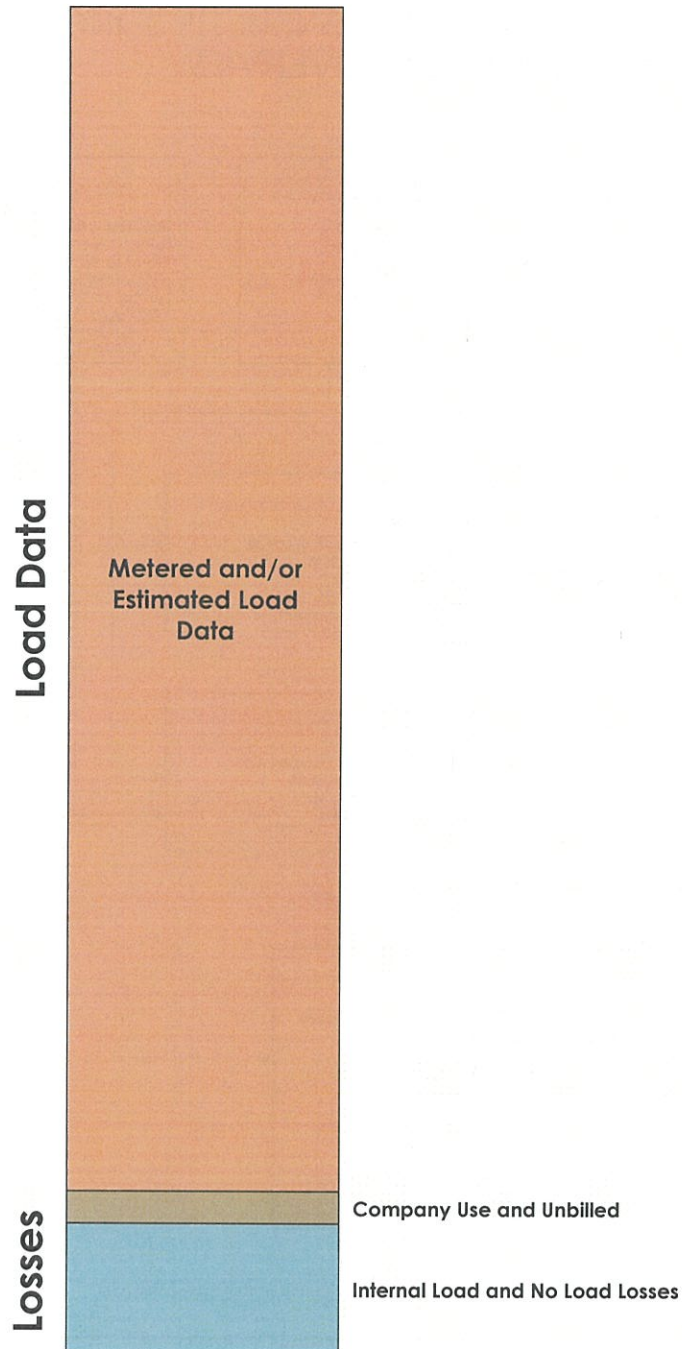
An overview of the loss study is shown on Figure 1 on the next page. Figure 2 simply illustrates the major components that must be considered in a loss analysis.

Oncor Electric Delivery System 2015 Analysis of System Losses



Oncor Electric Delivery System 2015 Analysis of System Losses

FIGURE 2
Generic Internal Energy Elements



Oncor Electric Delivery System 2015 Analysis of System Losses

2.0 INTRODUCTION

This report of the 2015 Analysis of System Losses for the Oncor Electric Delivery System provides a summary of results, conceptual background or methodology, description of the analyses, and input information related to the study.

2.1 Conduct of Study

Typically, between five to ten percent of the total peak hour MW and annual MWH requirements of an electric utility is lost or unaccounted for in the delivery of power to customers from the Transmission level through and including secondary voltage levels at the meter. Investments must be made in facilities which support the total load which includes losses or unaccounted for load. Revenue requirements associated with load losses are an important concern to utilities and regulators in that customers must equitably share in all of these cost responsibilities. Loss expansion factors by voltage level are the mechanism by which customers' metered demand and energy data are mathematically adjusted to the generation or input level (point of reference) when performing cost and revenue calculations.

An acceptable accounting of losses can be determined for any given time period using available engineering, system, and customer data along with empirical relationships. This loss analysis for the delivery of demand and energy utilizes such an approach. A microcomputer loss model³ is utilized as the vehicle to organize the available data, develop the relationships, calculate the losses, and provide an efficient and timely avenue for future updates and sensitivity analyses. Our procedures and calculations rely on numerous databases that include customer statistics and power system investments at various voltage levels of service.

Company personnel performed most of the data gathering and data processing efforts and checked for reasonableness. MAC provided assistance as necessary to construct databases, transfer files, perform calculations, and check the reasonableness of results. Efforts in determining the data required to perform the loss analysis centered on information which was available from existing studies or reports within the Company. From an overall perspective, our efforts concentrated on five major areas:

1. System information concerning ERCOT peak demand and annual energy requirements by voltage level of service using metered data and load research,
2. Substation loads at their input (including losses),
3. Distribution system primary and secondary loss calculations,
4. Derivation of fixed and variable losses by voltage level, and
5. Development of final cumulative expansion factors at each voltage for peak demand (kW) and annual energy (kWh) requirements at the point of delivery (meter).

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Oncor Electric Delivery System 2015 Analysis of System Losses

2.2 Electric Power Losses

Losses in power systems consist of primarily technical losses with a much smaller level of non-technical losses.

Technical Losses

Electrical losses result from the transmission of energy over various electrical equipment. The largest component of total losses during peaking conditions is power dissipation as a result of varying loading conditions and are oftentimes called load losses which are mostly related to the square of the current (I^2R). These peak hour losses can be as high as 60% to 75% of all technical losses during peak loading conditions. The remaining losses are called no-load and represent essentially fixed (constant) energy losses throughout the year. These no-load losses represent energy required to energize various electrical equipment regardless of their loading levels over the entire year. The major portion of these no-load losses consists of core or magnetizing energy related to installed transformers throughout the power system and generates the major component of annual losses on any distribution system.

The following Table 2 summarizes the unadjusted fixed and variable losses by major functional categories from Exhibit 5 of Appendix A:

TABLE 2

	<u>DEMAND (PEAK HOUR) – MW</u>			<u>ENERGY (ANNUAL AVERAGE) – MWH</u>		
	FIXED	VARIABLE	TOTAL	FIXED	VARIABLE	TOTAL
TRANS (%)	0.00	0.00	0.00 N/A	0	0	0 N/A
SUBTRANS (%)	0.00	0.00	0.00 N/A	0	0	0 N/A
DIST SUBS (%)	49.67 34.63%	93.76 65.37%	143.43 100.00%	341,427 56.21%	265,963 43.79%	607,390 100.00%
PRIMARY (%)	0.61 0.20%	298.94 99.80%	299.56 100.00%	5,350 0.57%	936,888 99.43%	942,238 100.00%
SECONDARY (%)	163.45 44.97%	199.98 55.03%	363.42 100.00%	1,431,800 74.84%	481,367 25.16%	1,913,167 100.00%
TOTAL SYS (%)	213.72 26.50%	592.68 73.50	806.40 100.00%	1,778,576 51.36%	1,684,217 48.64%	3,462,794 100.00%
TOTAL DIST (%)	213.72 26.50%	592.68 73.50%	806.40 100.00%	1,778,576 51.36%	1,684,217 48.64%	3,462,794 100.00%

Oncor Electric Delivery System 2015 Analysis of System Losses

Non-Technical Losses

These are unaccounted for energy losses that are related to energy theft, metering, non-payment by customers, and accounting errors. Losses related to these areas are generally very small and can be extremely difficult and subjective to quantify. Our efforts generally do not develop any meaningful level because we assume that improving technology and utility practices have minimized these amounts.

2.3 Loss Impacts from Distributed Generation (DG)

The impacts of losses on a power system from the installation of various DG facilities will depend somewhat on the penetration level, type of installations and location on a circuit. Based on the results presented in Table 2 of this loss study, the impacts are significantly different from looking at any single peak load hour versus the potential impacts over all hours of an entire year. Use of a typical uniform loss factor(s) for each voltage level may require additional consideration to recognize that a reduced consumption level could have little or no impact due to the recovery requirements for the high level of fixed losses over the entire hourly electric grid condition for any DG location.

2.4 Description of Model

The loss model is a customized applications model, constructed using the Excel software program. Documentation consists primarily of the model equations at each cell location. A significant advantage of such a model is that the actual formulas and their corresponding computed values at each cell of the model are immediately available to the analyst.

A brief description of the three (3) major categories of effort for the preparation of each loss model is as follows:

- Main sheet which contains calculations for all primary and secondary losses, summaries of all conductor and transformer calculations from other sheets discussed below, output reports and supporting results.
- Transformer sheet which contains data input and loss calculations for each distribution substation from high voltage transformer. Separate iron and winding losses are calculated for each transformer by identified type.
- Conductor sheet containing summary data by major voltage level as to circuit miles, loading assumptions, and kW and kWh loss calculations. This area of the model was not used as this loss study did not consider high voltage facilities.



Oncor Electric Delivery System 2015 Analysis of System Losses

3.0 METHODOLOGY

3.1 Background

The objective of a Loss Study is to provide a reasonable set of energy (average) and demand (peak) loss expansion factors which account for system losses associated with the transmission and delivery of power to each voltage level over a designated period of time. The focus of this study is to identify the difference between total energy inputs and the associated sales with the difference being equitably allocated to all delivery levels. Several key elements are important in establishing the methodology for calculating and reporting the Company's losses. These elements are:

- Selection of voltage level of services,
- Recognition of losses associated with conductors, transformations, and other electrical equipment/components within voltage levels,
- Identification of customers and loads at various voltage levels of service,
- Review of generation or net power supply input at each level for the test period studied, and
- Analysis of kW and kWh sales by voltage levels within the test period.

The three major areas of data gathering and calculations in the loss analysis were as follows:

1. System Information (monthly and annual)
 - MW at the distribution substations and metered MWH sales.
 - Coincident peak (ERCOT) estimates and net power supply input from all sources and voltage levels.
 - Customer load data estimates from available load research information, adjusted MWH sales, and number of customers in the customer groupings and voltage levels identified in the model.
 - System default values, such as power factor, loading factors, and load factors by voltage level.

Oncor Electric Delivery System 2015 Analysis of System Losses

2. Distribution System

- Distribution Substations – Data was developed for modeling each substation as to its size and loading. Loss calculations were prepared from this data to determine load and no load losses separately for each transformer.
- Primary lines – Line loading and loss characteristics for several representative primary circuits were obtained from the Company. These loss results developed kW loss per MW of load and a composite average was calculated to derive the primary loss estimate.
- Line transformers – Losses in line transformers were based on each customer service group's size, as well as the number of customers per transformer. Accounting and load data provided the foundation with which to model the transformer loadings and to calculate load and no load losses.
- Secondary circuits – Typical secondary circuits were estimated for conductor sizes, lengths, loadings, and customer penetration for residential and small general service customers based on data provided by the Company.
- Services – Typical services were estimated for each secondary service class of customers identified in the study with respect to type, length, and loading.
- Meters – Estimated one-hour losses were developed for each customer class and voltage level.

The loss analysis was thus performed by constructing the model in segments and subsequently calculating the composite until the constraints of peak demand and energy were met:

- Information as to the physical characteristics and loading of each transformer and conductor segment was modeled.
- Conductors, transformers, and distribution were grouped by voltage level, and unadjusted losses were calculated.
- The loss factors calculated at each voltage level were determined by “compounding” the per-unit losses. Equivalent sales at the supply point were obtained by dividing sales at a specific level by the compounded loss factor to determine losses by voltage level.



Oncor Electric Delivery System 2015 Analysis of System Losses

- The resulting demand and energy loss expansion factors were then used to adjust all sales to the generation or input level in order to estimate the difference.
- Reconciliation of kW and kWh sales by voltage level was accomplished by adjusting the initial loss factor estimates until the mismatch was eliminated.

3.2 Calculations and Analysis

This section provides a discussion of the input data, assumptions, and calculations performed in the loss analysis. Specific appendices have been included in order to provide documentation of the input data utilized in the model.

3.2.1 Transmission Systems

Not applicable.

3.2.2 Transformers

The transformer loss analysis required several steps in order to properly consider the characteristics associated with various transformer types; such as, distribution substations and line transformers. In addition, further efforts identified both iron and winding losses within each of these transformer types in order to obtain reasonable peak (kW) and average energy (kWh) losses. While iron losses were considered essentially constant for each hour, recognition had to be made for the varying degree of winding losses due to hourly equipment loadings.

Standardized test data tables were used to represent no load information (fixed) and full load (variable) losses for different sizes of transformers. This test data was incorporated into the loss model to develop relationships representing winding and iron or core losses for the transformer loss calculation as discussed in Section 4.0.

The remaining miscellaneous losses considered in the loss study consisted of several areas which do not lend themselves to any reasonable level of modeling for estimating their respective losses and were therefore lumped together into a single loss factor of 0.10%. The typical range of values for these losses is from 0.10% to 0.25%, and we have assumed the lower value to be conservative at this time. The losses associated with this loss factor include bus bars, unmetered station use, grounding transformers, cooling fans, heating and air conditioning requirements, and other remaining station use requirements.



Oncor Electric Delivery System 2015 Analysis of System Losses

3.2.3 Distribution System

The load data at the substation and customer level, coupled with primary and secondary network information, was sufficient to model the distribution system in adequate detail to calculate losses.

Primary Lines

Primary line loadings take into consideration the available distribution load along with the actual customer loads including losses. Primary line loss estimates were prepared by the Company for use in this loss study. These estimates considered loads per substation, voltage levels, loadings, total circuit miles, wire size, and single- to three-phase investment estimates. All of these factors were considered in calculating the actual demand (kW) and energy (kWh) for the primary system.

Line Transformers

Losses in line transformers were determined based on typical transformer sizes for each secondary customer service group and an estimated or calculated number of customers per transformer. Accounting records and estimates of load data provided the necessary database with which to model the loadings. These calculations also made it possible to determine separate winding and iron losses for distribution line transformers, based on a table of representative losses for various transformer sizes.

Secondary Line Circuits

A calculation of secondary line circuit losses was performed for loads served through these secondary line investments. Estimates of typical conductor sizes, lengths, loadings and customer class penetrations were made to obtain total circuit miles and losses for secondary. Customer loads which do not have secondary line requirements were also identified so that a reasonable estimate of losses and circuit miles of these investments could be made.

Service Drops and Meters

Service drops were estimated for each secondary customer reflecting conductor size, length and loadings to obtain demand losses. A separate calculation was also performed using customer maximum demands to obtain kWh losses. Meter loss estimates were also made for each customer and incorporated into the calculations of kW and kWh losses included in the Summary Results.



Oncor Electric Delivery System 2015 Analysis of System Losses

4.0 DISCUSSION OF RESULTS

A brief description of each Exhibit provided in Appendix A follows:

Exhibit 1 – Summary of Company Data

This exhibit reflects system information used to determine percent losses and a detailed summary of kW and kWh losses by voltage level. The loss factors developed in Exhibit 7 are also summarized by voltage level.

Exhibit 2 – Summary of Conductor Information

A summary of MW and MWH load and no load losses for conductors by voltage levels is presented. The sum of all calculated losses by voltage level is based on input data information provided in Appendix A. Percent losses are based on equipment loadings.

Exhibit 3 – Summary of Transformer Information

This exhibit summarizes transformer losses by various types and voltage levels throughout the system. Load losses reflect the winding portion of transformer losses while iron losses reflect the no load or constant losses. MWH losses are estimated using a calculated loss factor for winding and the test year hours times no load losses.

Exhibit 4 – Summary of Losses Diagram (2 Pages)

This loss diagram represents the inputs and output of power at system peak conditions. Page 1 details information from all points of the power system and what is provided to the distribution system for primary loads. This portion of the summary can be viewed as a “top down” summary into the distribution system.

Page 2 represents a summary of the development of primary line loads and distribution substations based on a “bottom up” approach. Basically, loadings are developed from the customer meter through the Company’s physical investments based on load research and other metered information by voltage level to arrive at MW and MVA requirements during peak load conditions by voltage levels.

Exhibit 5 – Summary of Sales and Calculated Losses

Summary of Calculated Losses represents a tabular summary of MW and MWH load and no load losses by discrete areas of delivery within each voltage level. Losses have been identified and are derived based on summaries obtained from Exhibits 2 and 3 and losses associated with meters, capacitors and regulators.



Oncor Electric Delivery System 2015 Analysis of System Losses

Exhibit 6 – Development of Loss Factors, Unadjusted

This exhibit calculates demand and energy losses and loss factors by specific voltage levels based on sales level requirements. The actual results reflect loads by level and summary totals of losses at that level, or up to that level, based on the results as shown in Exhibit 5. Finally, the estimated values at generation are developed and compared to actual generation to obtain any difference or mismatch.

Exhibit 7 – Development of Loss Factors, Adjusted

The adjusted loss factors are the results of adjusting Exhibit 6 for any difference. All differences between estimated and actual are prorated to each level based on the ratio of each level's total load plus losses to the system total. These new loss factors reflect an adjustment in losses due only to the kW and kWh mismatch.

Exhibit 8 – Adjusted Losses and Loss Factors by Facility

These calculations present an expanded summary detail of Exhibit 7 for each segment of the power system with respect to the flow of power and associated losses from the receipt of energy at the meter to the generation for the Oncor power system.

Exhibit 9 – Summary of Losses by Delivery Voltage

These calculations present a reformatted summary of losses presented in Exhibits 7 and 8 by power system delivery segment as calculated by voltage level of service based on reported metered sales.

**Oncor Electric Delivery System
2015 Analysis of System Losses**

Appendix A

**Results of 2015 Oncor Integrated
Power System Loss Analysis**



ONCOR 2015 LOSS ANALYSIS

ONCOR

EXHIBIT 1

SUMMARY OF COMPANY DATA

ONCOR ERCOT PEAK INPUT	22,679 MW
ANNUAL SYSTEM INPUT	106,435,000 MWH
ANNUAL SALES OUTPUT	102,972,872 MWH
SYSTEM LOSSES @ INPUT	3,462,128 or 3.25%
SYSTEM LOSSES @ OUTPUT	3,462,128 or 3.36%
SYSTEM LOAD FACTOR	53.6%

SUMMARY OF LOSSES - OUTPUT RESULTS

SERVICE	KV	--- MW --- Input	% TOTAL	--- MWH --- Input	% TOTAL
TRANS	345,138,69	0.0 0.00%	0.00%	0 0.00%	0.00%
PRIM SUBS	25 TO 4	143.5 0.63%	17.79%	607,273 0.57%	17.54%
PRIMARY	25 TO 4	299.7 1.32%	37.15%	942,056 0.89%	27.21%
SECONDARY	120/240,to,477	363.6 1.60%	45.07%	1,912,799 1.80%	55.25%
TOTAL		806.7 3.56%	100.00%	3,462,128 3.25%	100.00%

SUMMARY OF LOSS FACTORS

SERVICE	KV	CUMMULATIVE SALES EXPANSION FACTORS			
		DEMAND (Peak)		ENERGY (Annual)	
		d	1/d	e	1/e
TOT TRANS	345,138,69	1.00000	1.00000	1.00000	1.00000
PRIM SUBS	25 TO 4	1.00637	0.99367	1.00574	0.99429
PRIMARY	25 TO 4	1.02019	0.98021	1.01506	0.98517
SECONDARY	120/240,to,477	1.03885	0.96261	1.03717	0.96416

ONCOR 2015 LOSS ANALYSIS

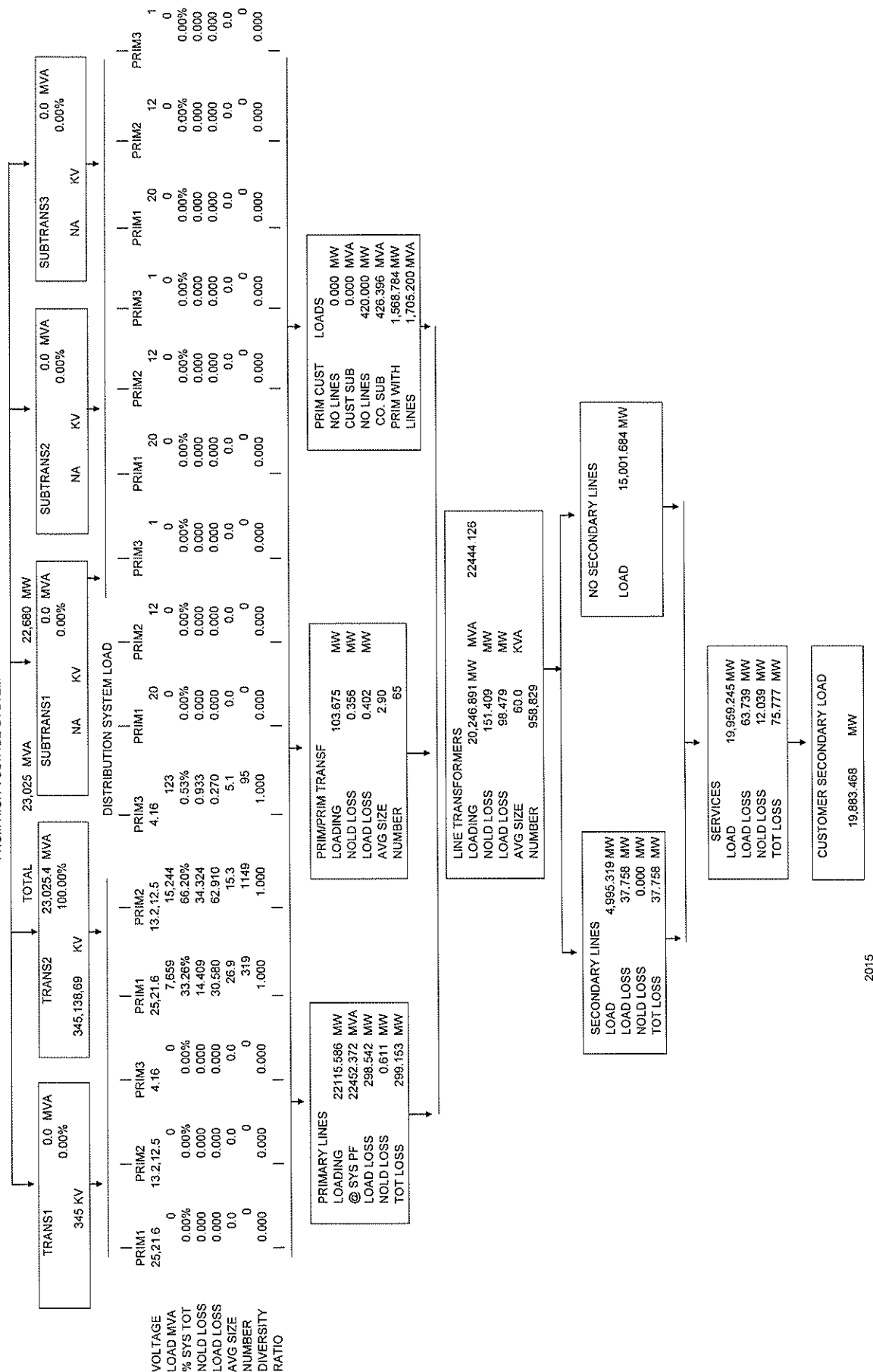
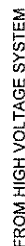
EXHIBIT 3

SUMMARY OF TRANSFORMER INFORMATION

DESCRIPTION	KV CAPACITY VOLTAGE	MVA	NUMBER TRANSFMR	AVERAGE SIZE	LOADING %	MVA LOAD	MW LOSSES		MW LOSSES		TOTAL	
							LOAD	NO LOAD	LOAD	NO LOAD	LOAD	TOTAL
BULK STEP-UP	NA	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
BULK - BULK	NA	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
BULK - TRANS1	NA	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
BULK - TRANS2	NA	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
TRANS1 STEP-UP	NA	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
TRANS1 - TRANS2	NA	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
TRANS1-SUBTRANS1	NA	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
TRANS1-SUBTRANS2	NA	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
TRANS1-SUBTRANS3	NA	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
TRANS2 STEP-UP	NA	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
TRANS2-SUBTRANS1	NA	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
TRANS2-SUBTRANS2	NA	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
TRANS2-SUBTRANS3	NA	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
SUBTRAN1 STEP-UP	NA	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
SUBTRAN2 STEP-UP	NA	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
SUBTRAN3 STEP-UP	NA	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
SUBTRAN1-SUBTRAN2	NA	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
SUBTRAN1-SUBTRAN3	NA	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
SUBTRAN2-SUBTRAN3	NA	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
DISTRIBUTION SUBSTATIONS												
TRANS1 -	25,21.6	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
TRANS1 -	13,2,12.5	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
TRANS1 -	4.16	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
TRANS2 -	25,21.6	8,592.0	319	26.9	89.14%	7,659	30,580	14,409	102,494	95,074	197,567	
TRANS2 -	13,2,12.5	17,575.0	1149	15.3	86.73%	15,244	62,910	34,324	162,770	238,681	401,451	
TRANS2 -	4.16	480.0	95	5.1	25.59%	123	0.270	0.933	699	7,672	8,371	
SUBTRAN1-	25,21.6	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
SUBTRAN1-	13,2,12.5	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
SUBTRAN1-	4.16	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
SUBTRAN2-	25,21.6	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
SUBTRAN2-	13,2,12.5	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
SUBTRAN2-	4.16	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
SUBTRAN3-	25,21.6	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
SUBTRAN3-	13,2,12.5	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
SUBTRAN3-	4.16	0.0	0	0.0	0.00%	0	0.000	0.000	0.000	0	0	0
PRIMARY - PRIMARY		188.5	65	2.9	55.00%	104	0.402	0.356	984	3,121	4,104	
LINE TRANSFMR		57,484.0	958,829	60.0	38.61%	22,194	98,479	151,409	222,724	1,326,342	1,549,066	
TOTAL		84,320	960,457				192,641	201,431	394,072	489,670	1,670,890	2,160,560



EXHIBIT 4 PAGE 2 of 2



ONCOR 2015 LOSS ANALYSIS

EXHIBIT 5

SUMMARY of SALES and CALCULATED LOSSES

LOSS # AND LEVEL	MW/LOAD	NO LOAD +	TOT LOSS	EXP FACTOR	CUM EXP FAC	MWH/LOAD	NO LOAD +	TOT LOSS	EXP FACTOR	CUM EXP FAC
1 BULK XFMR	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0
2 BULK LINES	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0
3 TRANS1 XFMR	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0
4 TRANS1 LINES	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0
5 TRANS2TR1 SD	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0
6 TRANS2BLK SD	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0
7 TRANS2 LINES	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0
8 STR1BLK SD	0.0	0.00	0.00	1.000000	1.000000	0	0	0	0	1.000000
9 STR1T1 SD	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0.000000
10 STR1T2 SD	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0.000000
11 SUBTRANS1 LINES	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0.000000
12 STR2T1 SD	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0.000000
13 STR2T2 SD	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0.000000
14 STR2S1 SD	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0.000000
15 SUBTRANS2 LINES	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0.000000
16 STR3T1 SD	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0.000000
17 STR3T2 SD	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0.000000
18 STR3S1 SD	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0.000000
19 STR3S2 SD	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0.000000
20 SUBTRANS3 LINES	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0.000000
21 SUBTRANS TOTAL	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0.000000
DISTRIBUTION SUBST										
TRANS1	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0.000000
TRANS2	22,678.7	49.67	143.43	1.006364	1.006364	106,435,666	341,427	265,963	607,390	1.0057394
SUBTR1	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0.000000
SUBTR2	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0.000000
SUBTR3	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0.000000
WEIGHTED AVERAGE	22,678.7	49.67	143.43	1.006364	1.006364	106,435,666	341,427	265,963	607,390	1.0057394
PRIMARY INTRCHGE	0.0	0.00	0.00	0.000000	0.000000	0	0	0	0	0.000000
PRIMARY LINES	22,115.2	0.61	299.56	1.013731	1.020183	102,610,160	5,350	936,888	942,238	1.0150604
LINE TRANSF	20,246.9	151.41	249.89	1.012496	1.032932	89,713,392	1,326,342	222,724	1,549,066	1.0175702
SECONDARY	19,997.0	0.00	37.76	1.001892	1.034886	88,164,325	0	87,940	87,940	1.0339265
SERVICES	19,959.2	12.04	63.74	1.003811	1.038830	88,076,386	105,458	170,703	276,161	1.0031453
TOTAL SYSTEM		213.72	592.68				1,778,576	1,684,217	3,462,794	

ONCOR 2015 LOSS ANALYSIS

DEVELOPMENT of LOSS FACTORS
UNADJUSTED
DEMAND

EXHIBIT 6

LOSS FACTOR LEVEL	CUSTOMER SALES MW a	CALC LOSS TO LEVEL b	SALES MW @ SUB INPUT c	CUM PEAK EXPANSION FACTORS d	1/d
BULK LINES	0.0	0.0	0.0	0.00000	0.00000
TRANS SUBS	0.0	0.0	0.0	0.00000	0.00000
TRANS LINES	0.0	0.0	0.0	1.00000	1.00000
TOTAL TRANS	0.0	0.0	0.0	0.00000	0.00000
SUBTRANS	0.0	0.0	0.0	1.00000	1.00000
PRIM SUBS	420.0	2.7	422.7	1.00636	0.99368
PRIM LINES	1,568.8	31.7	1,600.4	1.02018	0.98022
SECONDARY	<u>19,883.5</u>	<u>772.1</u>	<u>20,655.5</u>	1.03883	0.96262
TOTALS	21,872.3	806.4	22,678.7		

DEVELOPMENT of LOSS FACTORS
UNADJUSTED
ENERGY

LOSS FACTOR LEVEL	CUSTOMER SALES MWH a	CALC LOSS TO LEVEL b	SALES MWH @ SUB INPUT c	CUM ANNUAL EXPANSION FACTORS d	1/d
BULK LINES	0	0	0	0.00000	0.00000
TRANS SUBS	0	0	0	0.00000	0.00000
TRANS LINES	0	0	0	1.00000	1.00000
TOTAL TRANS	0	0	0	0.00000	0.00000
SUBTRANS	0	0	0	1.00000	1.00000
PRIM SUBS	3,218,116	18,470	3,236,586	1.00574	0.99429
PRIM LINES	11,954,531	180,040	12,134,571	1.01506	0.98516
SECONDARY	<u>87,800,225</u>	<u>3,264,284</u>	<u>91,064,509</u>	1.03718	0.96415
TOTALS	102,972,872	3,462,794	106,435,666		

ESTIMATED VALUES AT DIST SUB INPUT

LOSS FACTOR AT
VOLTAGE LEVEL

	MW	MWH
BULK LINES	0.00	0
TRANS SUBS	0.00	0
TRANS LINES	0.00	0
SUBTRANS SUBS	0.00	0
SUBTRANS LINES	0.00	0
PRIM SUBS	422.67	3,236,586
PRIM LINES	1,600.45	12,134,571
SECONDARY	20,655.54	91,064,509
SUBTOTAL	22,678.66	106,435,666
ACTUAL ENERGY	22,679.00	106,435,000
MISMATCH	(0.34)	666
% MISMATCH	0.00%	0.00%

ONCOR 2015 LOSS ANALYSIS

DEVELOPMENT of LOSS FACTORS

EXHIBIT 7

ADJUSTED
DEMAND

LOSS FACTOR LEVEL	CUSTOMER SALES MW a	SALES ADJUST b	CALC LOSS TO LEVEL c	SALES MW @ SUB INPUT d	CUM PEAK EXPANSION FACTORS e	f=1/e
BULK LINES	0.0	0.0	0.0	0.0	0.00000	0.00000
TRANS SUBS	0.0	0.0	0.0	0.0	0.00000	0.00000
TRANS LINES	0.0	0.0	0.0	0.0	1.00000	1.00000
TOTAL TRANS	0.0	0.0	0.0	0.0	0.00000	0.00000
SUBTRANS	0.0	0.0	0.0	0.0	1.00000	1.00000
PRIM SUBS	420.0	0.0	2.7	422.7	1.00637	0.99367
PRIM LINES	1,568.8	0.0	31.7	1,600.5	1.02019	0.98021
SECONDARY	<u>19,883.5</u>	<u>0.0</u>	<u>772.4</u>	<u>20,655.9</u>	1.03885	0.96261
			806.7			
TOTALS	21,872.3	0.0	806.7	22,679.0	1.03688	0.96443

DEVELOPMENT of LOSS FACTORS

ADJUSTED
ENERGY

LOSS FACTOR LEVEL	CUSTOMER SALES MWH a	SALES ADJUST b	CALC LOSS TO LEVEL c	SALES MWH @ SUB INPUT d	CUM ANNUAL EXPANSION FACTORS e	f=1/e
BULK LINES	0	0	0	0	0.00000	0.00000
TRANS SUBS	0	0	0	0	0.00000	0.00000
TRANS LINES	0	0	0	0	1.00000	1.00000
TOTAL TRANS	0	0	0	0	0.00000	0.00000
SUBTRANS	0	0	0	0	1.00000	1.00000
PRIM SUBS	3,218,116	0	18,466	3,236,582	1.00574	0.99429
PRIM LINES	11,954,531	0	180,005	12,134,536	1.01506	0.98517
SECONDARY	<u>87,800,225</u>	<u>0</u>	<u>3,263,656</u>	<u>91,063,881</u>	1.03717	0.96416
			3,462,128			
TOTALS	102,972,872	0	3,462,128	106,435,000	1.03362	0.96747

ESTIMATED VALUES AT DIST SUB INPUT

LOSS FACTOR AT
VOLTAGE LEVEL

	MW	MWH
BULK LINES	0.00	0
TRANS SUBS	0.00	0
TRANS LINES	0.00	0
SUBTRANS SUBS	0.00	0
SUBTRANS LINES	0.00	0
PRIM SUBS	422.67	3,236,582
PRIM LINES	1,600.46	12,134,536
SECONDARY	20,655.86	91,063,881
	22,679.00	106,435,000
ACTUAL ENERGY	22,679.00	106,435,000
MISMATCH	(0.00)	(0)
% MISMATCH	0.00%	0.00%

ONCOR 2015 LOSS ANALYSIS

Adjusted Losses and Loss Factors by Facility

EXHIBIT 6

Unadjusted Losses by Segment				
	MW	Unadjusted	MWH	Unadjusted
Service Drop Losses	75.78	75.78	276,161	276,161
Secondary Losses	37.76	37.76	87,940	87,940
Line Transformer Losses	249.89	249.89	1,549,066	1,549,066
Primary Line Losses	299.56	299.56	942,238	942,238
Distribution Substation Losses	143.43	143.43	607,390	607,390
Subtransmission Losses	0.00	0.00	0	0
<u>Transmission System Losses</u>	<u>0.00</u>	<u>0.00</u>	<u>0</u>	<u>0</u>
Total	806.40	806.40	3,462,794	3,462,794
Mismatch Allocation by Segment				
	MW		MWH	Note adjusting
Service Drop Losses	-0.03		53	53
Secondary Losses	-0.02		17	17
Line Transformer Losses	-0.11		298	298
Primary Line Losses	-0.13		181	181
Distribution Substation Losses	-0.06		117	117
Subtransmission Losses	0.00		0	0
<u>Transmission System Losses</u>	<u>0.00</u>		<u>0</u>	<u>0</u>
Total	-0.34		666	666
Adjusted Losses by Segment				
	MW	% of Total	MWH	% of Total
Service Drop Losses	75.81	9.4%	276,108	8.0%
Secondary Losses	37.77	4.7%	87,923	2.5%
Line Transformer Losses	249.99	31.0%	1,548,768	44.7%
Primary Line Losses	299.68	37.1%	942,056	27.2%
Distribution Substation Losses	143.49	17.8%	607,273	17.5%
Subtransmission Losses	0.00	0.0%	0	0.0%
<u>Transmission System Losses</u>	<u>0.00</u>	<u>0.0%</u>	<u>0</u>	<u>0.0%</u>
Total	806.75	100.0%	3,462,128	100.0%
Loss Factors by Segment				
	MW		MWH	
Retail Sales from Service Drops	19,883.47		87,800,225	
<u>Adjusted Service Drop Losses</u>	<u>75.81</u>		<u>276,108</u>	
Input to Service Drops	19,959.28		88,076,333	
Service Drop Loss Factor	1.00381		1.00314	
Output from Secondary	19,959.28		88,076,333	
<u>Adjusted Secondary Losses</u>	<u>37.77</u>		<u>87,923</u>	
Input to Secondary	19,997.05		88,164,255	
Secondary Conductor Loss Factor	1.00189		1.00100	
Output from Line Transformers	19,997.05		88,164,255	
<u>Adjusted Line Transformer Losses</u>	<u>249.99</u>		<u>1,548,768</u>	
Input to Line Transformers	20,247.05		89,713,024	
Line Transformer Loss Factor	1.01250		1.01757	
Secondary Composite	1.01829		1.02179	
Retail Sales from Primary	1,454.78		11,494,341	
Req. Whls Sales from Primary	114.00		460,190	
<u>Input to Line Transformers</u>	<u>20,247.05</u>		<u>89,713,024</u>	
Output from Primary Lines	21,815.83		101,667,555	
<u>Adjusted Primary Line Losses</u>	<u>299.68</u>		<u>942,056</u>	
Input to Primary Lines	22,115.51		102,609,611	
Primary Line Loss Factor	1.01374		1.00927	
Output PI from Distribution Substations	22,115.51		102,609,611	
Req. Whls Sales from Substations	73.00		305,471	
Retail Sales from Substations	347.00		2,912,645	
Total Output from Distribution Substations	22,535.51		105,827,727	
<u>Adjusted Distribution Substation Losses</u>	<u>143.49</u>		<u>607,273</u>	
Input to Distribution Substations	22,679.00		106,435,000	
Distribution Substation Loss Factor	1.00637		1.00574	
Retail Sales at from SubTransmission	0.00		0	
Req. Whls Sales from SubTransmission	0.00		0	
<u>Input to Distribution Substations</u>	<u>0.00</u>		<u>0</u>	
Output from SubTransmission	0.00		0	
<u>Adjusted SubTransmission System Losses</u>	<u>0.00</u>		<u>0</u>	
Input to SubTransmission	0.00		0	
SubTransmission Loss Factor	1.00000		1.00000	
<u>Input to Distribution Substations</u>	<u>22,678.66</u>		<u>106,435,666</u>	
Retail Sales at from Transmission	0.00		0	
Req. Whls Sales from Transmission	0.00		0	
<u>Input Subtransmission</u>	<u>0.00</u>		<u>0</u>	
Output from Transmission	22,679.00		106,435,000	
<u>Adjusted Transmission System Losses</u>	<u>0.00</u>		<u>0</u>	
Input to Transmission	22,679.00		106,435,000	
Transmission Loss Factor	1.00000		1.00000	

DEMAND MW			SUMMARY OF LOSSES AND LOSS FACTORS BY DELIVERY VOLTAGE					EXHIBIT 9 PAGE 1 of 2
	SERVICE LEVEL	SALES MW	LOSSES	SECONDARY	PRIMARY	SUBSTATION		
1	SERVICES							
2	SALES	19,883.47		19,883.5				
3	LOSSES		75.8	75.8				
4	INPUT			19,959.3				
5	EXPANSION FACTOR	1.00381						
6	SECONDARY							
7	SALES							
8	LOSSES		37.8	37.8				
9	INPUT			19,997.1				
10	EXPANSION FACTOR	1.00189						
11	LINE TRANSFORMER							
12	SALES							
13	LOSSES		250.0	250.0				
14	INPUT			20,247.0				
15	EXPANSION FACTOR	1.01250						
16	PRIMARY							
17	SECONDARY			20,247.0				
18	SALES	1568.78			1,568.8			
19	LOSSES		299.7	278.1	21.6			
20	INPUT							
21	EXPANSION FACTOR	1.01374						
22	SUBSTATION							
23	PRIMARY			20,525.2	1,590.3			
24	SALES	420.0				420.0		
25	LOSSES		143.5	130.7	10.1	2.7		
26	INPUT			20,655.9	1,600.5	422.7		
27	EXPANSION FACTOR	1.00637						
28	SUB-TRANSMISSION							
29	DISTRIBUTION SUBS			17,309.6				
30	SALES	0.00			832.2	0.0		
31	LOSSES		0.0	0.0	0.0	0.0		
32	INPUT			17,309.6	832.2	0.0		
33	EXPANSION FACTOR	1.00000						
34	TRANSMISSION							
35	SUBTRANSMISSION			17,309.6	832.2			
36	DISTRIBUTION SUBS			3,346.3	768.2	422.7		
37	SALES	0.00						
38	LOSSES		0.0	0.0	0.0	0.0		
39	INPUT			20,655.9	1,600.5	422.7		
40	EXPANSION FACTOR	1.00000						
41	TOTALS	LOSSES	CALCULATED	806.7	772.4	31.7	2.7	
			SCALED	806.7	772.4	31.7	2.7	
42		% OF TOTAL		100%	95.74%	3.93%	0.33%	
43		SALES	21,872.3		19,883.5	1,568.8	420.0	
44		% OF TOTAL	100.00%		90.91%	7.17%	1.92%	
45		INPUT	22,679.0		20,655.9	1,600.5	422.7	
46	CUMMULATIVE EXPANSION LOSS FACTORS (from motor to system input)		1.03688 SYSTEM	1.03885	1.02019	NA		

ENERGY MWH			SUMMARY OF LOSSES AND LOSS FACTORS BY DELIVERY VOLTAGE				EXHIBIT 9
SERVICE LEVEL			SALES	LOSSES	SECONDARY	PRIMARY	SUBSTATION
PAGE 2 of 2							
1	SERVICES						
2	SALES		87,800,225		87,800,225		
3	LOSSES			276,108	276,108		
4	INPUT				88,076,333		
5	EXPANSION FACTOR	1.00314					
6	SECONDARY						
7	SALES			87,923	87,923		
8	LOSSES				88,164,255		
9	INPUT						
10	EXPANSION FACTOR	1.00100					
11	LINE TRANSFORMER						
12	SALES			1,548,768	1,548,768		
13	LOSSES				89,713,024		
14	INPUT						
15	EXPANSION FACTOR	1.01757					
16	PRIMARY						
17	SECONDARY				89,713,024		
18	SALES		11,954,531.000			11,954,531	
19	LOSSES			942,056	831,285	110,771	
20	INPUT						
21	EXPANSION FACTOR	1.00927					
22	SUBSTATION						
23	PRIMARY				90,544,309	12,065,302	
24	SALES		3,218,116				3,218,116
25	LOSSES			607,273	519,572	69,234	18,467
26	INPUT				91,063,881	12,134,537	3,236,583
27	EXPANSION FACTOR	1.00574					
28	SUB-TRANSMISSION						
29	DISTRIBUTION SUBS				81,320,045	4,805,277	0.000
30	SALES		0				0.000
31	LOSSES			0	0	0	0.000
32	INPUT				81,320,045	4,805,277	
33	EXPANSION FACTOR	1.00000					
34	TRANSMISSION						
35	SUBTRANSMISSION				81,320,045	4,805,277	
36	DISTRIBUTION SUBS				9,743,835	7,329,260	3,236,583
37	SALES		0				
38	LOSSES			0	0	0	0
39	INPUT				91,063,881	12,134,537	3,236,583
40	EXPANSION FACTOR	1.00000					
41	TOTALS	LOSSES	Calculated	3,462,128	3,263,656	180,006	18,467
			Scaled	3,462,128	3,263,656	180,005	18,467
42	% OF TOTAL			100%	94.27%	5.20%	0.53%
43	SALES		102,972,872		87,800,225	11,954,531	3,218,116
44	% OF TOTAL		100.00%		85.27%	11.61%	3.13%
45	INPUT		106,435,000		91,063,881	12,134,536	3,236,583
46	CUMMULATIVE EXPANSION LOSS FACTORS			1.03362	1.03717	1.01506	NA
	(from meter to system input)			SYSTEM			

**Oncor Electric Delivery System
2015 Analysis of System Losses**

Appendix B

Discussion of Hoebel Coefficient



Oncor Electric Delivery System 2015 Analysis of System Losses

COMMENTS ON THE HOEBEL COEFFICIENT

The Hoebel constant represents an established industry standard relationship between peak losses and average losses and is used in a loss study to estimate energy losses from peak demand losses. H. F. Hoebel described this relationship in his article, "Cost of Electric Distribution Losses," Electric Light and Power, March 15, 1959.

Within any loss evaluation study, peak demand losses can readily be calculated given equipment resistance and approximate loading. Energy losses, however, are much more difficult to determine given their time-varying nature. This difficulty can be reduced by the use of an equation which relates peak load losses (demand) to average losses (energy). Once the relationship between peak and average losses is known, average losses can be estimated from the known peak load losses.

Within the electric utility industry, the relationship between peak and average losses is known as the loss factor. For definitional purposes, loss factor is the ratio of the average power loss to the peak load power loss, during a specified period of time. This relationship is expressed mathematically as follows:

$$\underline{(1) F_{LS} \cong A_{LS} \div P_{LS}}$$

where: F_{LS} = Loss Factor
 A_{LS} = Average Losses
 P_{LS} = Peak Losses

The loss factor provides an estimate of the degree to which the load loss is maintained throughout the period in which the loss is being considered. In other words, loss factor is the ratio of the actual kWh losses incurred to the kWh losses which would have occurred if full load had continued throughout the period under study.

Examining the loss factor expression in light of a similar expression for load factor indicates a high degree of similarity. The mathematical expression for load factor is as follows:

$$\underline{(2) F_{LD} \cong A_{LD} \div P_{LD}}$$

where: F_{LD} = Load Factor
 A_{LD} = Average Load
 P_{LD} = Peak Load

This load factor result provides an estimate of the degree to which the load loss is maintained throughout the period in which the load is being considered. Because of the similarities in definition, the loss factor is sometimes called the "load factor of losses." While the definitions are similar, a strict equating of the two factors cannot be made. There does exist, however, a relationship between these two factors which is dependent upon the shape of the load duration curve. Since resistive losses vary as the square of the load, it can be shown mathematically that the loss factor can vary between the extreme limits of load factor and load factor squared. The

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relationship between load factor and loss factor has become an industry standard and is as follows:

$$(3) \ F_{LS} \cong H * F_{LD}^2 + (1-H) * F_{LD}$$

where: F_{LS} = Loss Factor
 F_{LD} = Load Factor
 H = Hoebel Coefficient

As noted in the attached article, the suggested value for H (the Hoebel coefficient) is 0.7. The exact value of H will vary as a function of the shape of the utility's load duration curve. In recent years, values of H have been computed directly for a number of utilities based on EEL load data. It appears on this basis, the suggested value of 0.7 should be considered a lower bound and that values approaching unity may be considered a reasonable upper bound. Based on experience, values of H have ranged from approximately 0.85 to 0.95. The standard default value of 0.9 is generally used.

Inserting the Hoebel coefficient estimate gives the following loss factor relationship using Equation (3):

$$(4) \ F_{LS} \cong 0.90 * F_{LD}^2 + 0.10 * F_{LD}$$

Once the Hoebel constant has been estimated and the load factor and peak losses associated with a piece of equipment have been estimated, one can calculate the average, or energy losses as follows:

$$(5) \ A_{LS} \cong P_{LS} * [H * F_{LD}^2 + (1-H) * F_{LD}]$$

where: A_{LS} = Average Losses
 P_{LS} = Peak Losses
 H = Hoebel Coefficient
 F_{LD} = Load Factor

Loss studies use this equation to calculate energy losses at each major voltage level in the analysis.